stability across cycles and up to 300°C, establishing new design principles and promising all-solid-state energy storage with rapid charging, extended lifespan, and high stability. Dielectric capacitors provide intense pulsed energy for various applications such as hybrid vehicles, accelerators, and microwave devices.

Organic electronics: Sustainable lifecycle approach

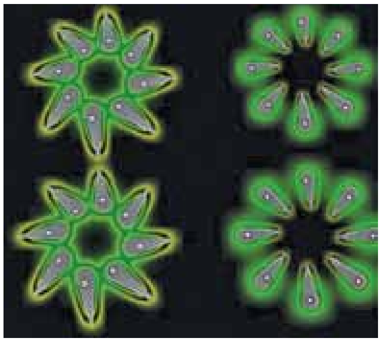
Scientists from Friedrich-Alexander University (FAU), along with researchers from the UK and the US, have proposed a circular strategy in a recent publication in *Nature Materials* (June 2023), highlighting the sustainable approach towards organic electronics throughout their entire lifecycle. Organic electronics, such as solar modules, offer flexibility and diverse applications, reducing dependence on rare materials like iridium, platinum, and silver. The development of organic electronics focuses on greener methods, including water-based deposition and inkjet printing, which contribute to reduced energy consumption and cost efficiency. Addressing challenges such as non-toxic solvents, durable materials, and the replacement of precious metals in OLED screens, organic electronics contribute to energy and resource conservation. To assess the environmental impact of organic electronics, a holistic view of the entire lifecycle is crucial. For instance, organic photovoltaic systems emit 30% less carbon-dioxide during manufacturing compared to silicon modules.

Saltwater: The future of green hydrogen production

Scientists from the Department of Physics at the Indian Institute of Technology (IIT) Madras have successfully developed a low-cost prototype that harnesses the power of seawater to generate green hydrogen (GH2). Currently, India relies on six million tonnes of grey hydrogen derived from natural gas, but transitioning to green hydrogen would require 132-192 million tonnes of water. Researchers worldwide are exploring desalination and direct seawater usage as means of producing GH2. While desalination yields clean water suitable for GH2 production, the process of seawater splitting poses significant challenges. Renewable-powered electrolysis has been utilised, but the high costs of desalination hinder efficient electrolysis. Although the technology readiness level (TRL) of seawater-based green hydrogen production is currently low, ongoing efforts are advancing its TRL and evaluating its techno-economic viability. Commercial viability may take several years to determine, but the potential of seawater hydrogen production is promising. The technology also encompasses the exploration of wastewater utilisation, including experiments with industrial wastewater.

Nanoscale integration: Precise growth of halide perovskite nanocrystals

MIT researchers have developed a groundbreaking technique for the precise growth of individual halide perovskite nanocrystals, allowing for controlled placement with a range of less than 50 nanometres. Traditional methods of integrating halide perovskites into nanoscale devices involved using fragile films with lithography or placing smaller crystals, but these approaches had limited control, hindering the extension of nanodevices. The new technique involves the direct growth of perovskite crystals onto surfaces using a template with wells designed for precise nanocrystal growth. By adjusting the well size and growth solution volume, the researchers achieved high precision in growth, positioning, and control of crystal size. The resulting nanoLED arrays have applications in optical communication, computing, quantum light sources, microscopy, and high-resolution displays for augmented and virtual reality.



A new MIT platform enables researchers to grow halide perovskite nanocrystals with precise control over the location and size of each individual crystal, integrating them into nanoscale light-emitting diodes. Pictured is a rendering of a nanocrystal array emitting light (Credit: Sampson Wilcox, RLE)

Magnetic robots unlocking boundless possibilities

Scientists at MIT have revolutionised the field of robotics with the creation of miniature, flexible robots embedded with rubbery magnetic spirals. These remarkable soft-bodied robots respond to weak magnetic fields and exhibit impressive versatility in their ability to walk, crawl, and swim. By strategically magnetising the robots' flexible bodies, they can be effortlessly controlled by a single magnetic field. Through the stretching of two rubber materials and the application of specific magnetisation patterns, these robots are capable of a diverse range of movements. Each robot is precisely magnetised, allowing for easy manipulation through the activation of a weak magnetic field. Remarkably, a single magnetic field can induce opposite movements in multiple robots, akin to flipping a switch. Such adjustability enables delicate payload release, which has significant implications for various applications. The researchers envision the application of these soft-bodied robots in the delivery of materials within pipes or even within the human body, reminiscent of drug transport in blood vessels.